

Research Proposal

Shifted Moving Block Bootstrap for Granger Causal Discovery

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1 Background

1.1 Predictive Causality

Causal discovery or causal inference in time series analysis can be thought of as the study of inferring causal links given observed time series data. In a bivariate setting we may want to know whether there is evidence to suggest X “causes” Y (denoted as $X \rightarrow Y$). Structural causality is largely divorced from the fundamentals of mainstream statistics ever since Francis Galton’s journey to formalise causality in hereditary genetics ironically resulted in his formalisation of correlation; as a result, definitions of causality can be grouped as either predictive or structural, with the former easier to define under statistical axioms [1].

The first notable attempt to define predictive causality in a temporal sense is Granger causality (GC), where having observed $X = \{X_i\}_{i=1}^n$ and $Y = \{Y_i\}_{i=1}^n$, X is said to Granger cause Y if $\text{Var}(\hat{Y}_{t+1} \mid \mathcal{H}_t) < \text{Var}(\hat{Y}_{t+1} \mid \mathcal{H}_t/X)$ where $\mathcal{H}_t$ is the full history of the system at time $t \in \{1, \dots, n\}$ [2]. Informally, X Granger causes Y if observed realisations of $X_{1:t-1}$ reduce the variance of the linear prediction of Y_t . Other measures include transfer entropy, which is a measure of information transfer rather than forecast variance [3], and Convergent Cross Mapping is a popular method based on coupling and constructing a manifold to map the causal structure [4],[5]. These measures can be labelled as predictive as we cannot discern whether true causality is taking place, or rather latent correlation through confounding variables.

1.2 Granger Causality Tests

Granger causality remains one of the most popular methods for quantifying predictive causality, with ready applications in neuroscience, econometrics and climate science due to its intuitive definition and computational efficiency [6]. Improving the accuracy of techniques for inferring the presence or absence of GC is therefore of great importance to researchers using GC to evaluate hypotheses.

To test the hypothesis of GC under a vector autoregressive (VAR) model we can carry out an F -test comparing the residuals of forecasts under the null and under the alternative hypothesis. Under stationarity, correct model specification and normality, this will output a test statistic following an F -distribution under the null, allowing us to compute a quantile and hence a p-value. Conversely, we can relax the Gaussian assumption and calculate a Wald statistic, though this needs a comparatively large sample size to reliably estimate a chi-squared quantile. Crucially, both of these tests require an estimated maximum lag of causal effect p which we will leverage in our proposed method. More detail on these tests is given in the supplementary material.

The objective of incorporating bootstrapping into GC tests is to relax distributional assumptions from these tests and interpret the test statistic in a non-parametric sense. By creating bootstrap samples, we seek to create synthetic realisations of our time series under H_0 , re-calculate the test statistic and see what proportion of bootstrap samples exhibit a GC test statistic (or any other causal measure) as extreme as observed on the original dataset. Clearly, to avoid Type II error we want the bootstrapping procedure to effectively break up any causality present without breaking up the within-chain autocorrelation structure to create accurate samples under H_0 .

Efron’s classical bootstrap has been shown to be asymptotically inconsistent for correlated data [7], and this motivated the development of block or window bootstrap methods which seek to preserve the autocorrelation structure of correlated data. Existing work on improving the accuracy of GC test using these bootstrapping procedures is currently focused on spectral transformations or randomised variations on the moving block bootstrap. [8], [9]. Whilst Farnè M and Montanari’s **grangers** [10] package in R proves to be a comprehensive tool for bootstrapping GC tests in the frequency domain, resampling schemes for GC tests in the time domain lack features for breaking up the underlying causality structure of the data, leading to underpowered tests.

1.3 Shifted Moving Block Bootstrap Hypothesis Test

Proposed by Yin et al. [11], the shifted moving block bootstrap hypothesis test is a method developed for extremal causal discovery, where causality occurs primarily in the tails in both cause and effect variables. Given a bivariate time series $X = \{X_i\}_{i=1}^n, Y = \{Y_i\}_{i=1}^n$, we can construct the following hypotheses:

$$H_0 = X \not\rightarrow Y, \quad H_1 = X \rightarrow Y$$

Given an estimated[†] lag p , some function $f(\cdot)$ and a chosen causal measure $\theta_{X \rightarrow Y}(f(p))$, we can calculate our benchmark statistic on X, Y . What to set $f(\cdot)$ to was left as an open question by Yin et al. [11], though we can consider $f(x) = x$ as a standard approach.

We then perform a circular shifting step, where we set $Y'_t = Y_{(t-f(p)) \bmod n}, \forall t \in \{1, \dots, n\}$ and denote $Y' = \{Y'_i\}_{i=1}^n$, then perform a moving block bootstrap (MBB) on X, Y' such that $(X, Y') \mapsto (X_*, Y'_*)$, repeating B times and calculating the proportion of bootstrapped $\theta_{X_* \rightarrow Y'_*}^b(f(p)), b \in \{1, \dots, B\}$ more extreme than our original $\theta_{X \rightarrow Y}(f(p))$ giving us a p-value informing us to reject or accept H_0 . Yin et al. [11] showed empirical findings on the shifted MBB outperforming a permutation testing approach for a particular $\theta_{X \rightarrow Y}(f(p))$ for detecting extremal causality, as well as its asymptotic consistency. Although comparison with the permutation test highlighted the benefit of the MBB approach, there was little empirical evidence of the impact of the shifting step.

As we must use the estimated maximal causal lag p as a hyperparameter in any GC testing, it seems intuitive to leverage this information in breaking up the causality structure (as this procedure does) yet few methods in the literature do. Misspecification of this parameter was shown to have a significant impact on both the size and power of the test in the case of the extremal causal measure (See Yin et al. S2.3 [11]). The optimisation of $f(p)$ as well as the optimal length of blocks used in MBB are potential extensions for the project.

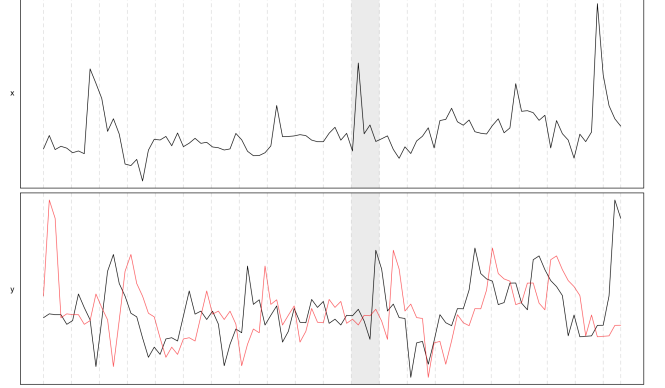


Figure 1: Top plot: realisation of X . Bottom plot: black line is realisation of Y , red line of Y' . Ground truth is $X \rightarrow Y$. Dashed lines are example block locations. Highlighted block shows the disruption of intra-block causal structure from shift.

2 Research Objectives

2.1 Primary Objectives

This research project will seek to extend the bivariate shifted block bootstrap hypothesis test to inferring Granger causality and record its empirical and theoretical performance.

Given synthetic bivariate data with a known ground truth causal structure of $X \rightarrow Y$ and a known time series model, we will repeat the shifted moving block bootstrap hypothesis test N times and calculate the proportion of significant p-values. This gives our proportion of correct inference in $X \rightarrow Y$ direction (true positives). Repeating this in the $Y \rightarrow X$ direction, we take the proportion of p-values which are insignificant as our proportion of correct inference (true negatives). The optimisation of these is analogous to maximising power and reducing size of the test.

We will then compare these rates to the asymptotic tests and existing bootstrapping methods. We will repeat this for a range of time series models, with a range of causal interactions and noise distributions. For models with encouraging empirical results, we will find the asymptotic distribution of the bootstrapped statistics given these model forms, hence attempting to prove the validity of this procedure under certain assumptions.

2.2 Extensions

Further directions to consider (ordered by feasibility) include:

- Exploring the optimisation and mathematical properties of bootstrap shift $f(p)$ hyperparameter and its relation to block length.
- Improving asymptotic proof of bootstrapping procedure by bounding convergence.
- Extending framework to real-world data. For instance, causal time series data with known ground truth can be found on the CauseMe [12] platform. This would shed light on the practical utility (or lack thereof) of the methodology.

[†]Assume p to be well estimated: existing methods are similar to estimating the lag in a parametric time series model, for example via AIC/BIC.

- Testing for confounding variables with multivariate GC.
- Exploring application for other predictive causality methods. This may be challenging as a benefit of GC is its efficiency to calculate, carrying out a bootstrapping procedure for a measure like transfer entropy will be very computationally expensive.

2.3 Contingency Planning

Both the simulation study and the mathematical proofs from Section 2.1 are judged to be feasible given the allotted time. A deconstructed view of the timeline is given in Section 4. However, it is wise to propose alternative primary objectives in the case that our feasibility assessment is misjudged.

We propose two alternative objectives, firstly the optimisation of the shift and block length hyperparameters. Yin et al. [11] showed the shifted MBB outperforms existing methods for extremal compound causality, but details on hyperparameter input is imperative for its practical use as a causal discovery measure. We could drop the GC angle and investigate hyperparameter optimisation for block length and the shift parameter on the compound causal tail coefficient detailed in the paper.

For our second alternative, our review of GC literature will give us a strong starting point for a comparison-based simulation model on current methods of GC discovery. Choosing a suitable parametric time series model, we can evaluate the performance of existing methods and apply them to real-world data which current research has neglected[†].

3 Methodology

3.1 Software

Simulations, visualisations and empirical analyses will be carried out in R. Yin et al. [11] provide skeleton code for the shifted MBB hypothesis test framework which will be used as a basis for analyses and both the Wald and F -tests can be carried out efficiently under the `lmtest` package.

3.2 Time Series Models

The analyses require suitable parametric models in order to evaluate the effectiveness of our proposed method. Under a stationary VAR model with Gaussian noise and a large number of samples we cannot expect to outperform the F -test, it is UMP for maximal causal lag $p = 1$ and UMP invariant for $p > 1$ (For a proof of maximal invariance and UMP condition see Lehmann and Romano: Sections 7.1,7.2 [13]). We therefore must change either the noise structure or the dependence structure of our parametric model. NAAR and GARCH are strong candidates; preliminary results show promise for a NAAR model with a sublinear causal structure. More detail on these models as well as results are given in the supplementary material.

3.3 Asymptotic Proof

Speculating on the structure of proving the validity of this procedure and pre-empting potential difficulties is inherently challenging. Ideally our test will have a p -value tending to 0 as we increase sample size and bootstrap replications for the case of H_1 being true, and a uniformly distributed p -value for the case that H_0 is true. The resampling nature of a bootstrap estimator is such that proving its consistency often relies on a WLLN-adjacent argument, so we can expect a successful proof to show convergence in probability under certain assumptions. From preliminary results and existing literature we expect these assumptions to include stationarity and strong mixing restrictions on the two time series processes [11].

4 Research Timeline

Table 1: Project Timeline

Task	Mar		Apr		May		Jun		Jul		Aug	
Literature review												
Preliminary simulations												
Examination period												
Comparative simulations on GC (2.1)												
Mathematical arguments (2.1,2.2)												
Report writing												
Research poster preparation												
Further simulation studies (2.2)												
Oral presentation preparation												

[†]If this route were taken, it would require additional reading to find gaps in data types explored by existing literature.

S1 Supplementary Material

S1.1 Asymptotic GC tests

Under the null hypothesis, assuming a VAR model we can perform the following tests:

$$F = \frac{(\sum_{t=1}^n \hat{\epsilon}_{R,t}^2 - \sum_{t=1}^n \hat{\epsilon}_{U,t}^2)/p}{\sum_{t=1}^n \hat{\epsilon}_{U,t}^2/(n-2p-1)} \sim F_{p,n-2p-1}, \quad W = \hat{\gamma}' [\text{Cov}(\hat{\gamma})]^{-1} \hat{\gamma} \sim \chi_p^2$$

Where p is the maximum causal lag hypothesised, $\hat{\gamma}$ is the estimated linear causal coefficient vector, R is the null (restricted) model exhibiting no causality, U is the alternative (unrestricted) model exhibiting causality and $\hat{\epsilon}$ is the residual sum of squares from the linear predictions. The distributional assumptions will hold where the underlying model is a stationary VAR, and in the F case where the white noise structure is Gaussian. Note that Wald's W does generally outperform the F -test for heavy-tailed noise but is generally not robust for very heavy tails and struggles with smaller samples. Both will perform poorly with non-linear causal dependencies. For more information see Shojaie A and Fox EB [6].

S1.2 VAR/NAAR

A non-linear additive auto regressive (NAAR) model is of the form:

$$\begin{bmatrix} X_t \\ Y_t \end{bmatrix} = \begin{bmatrix} \alpha_1 \\ \alpha_2 \end{bmatrix} + \sum_{i=1}^q \begin{bmatrix} f_{11,i}(X_{t-i}) + f_{12,i}(Y_{t-i}) \\ f_{21,i}(X_{t-i}) + f_{22,i}(Y_{t-i}) \end{bmatrix} + \begin{bmatrix} \epsilon_{X,t} \\ \epsilon_{Y,t} \end{bmatrix}$$

Where $\epsilon_{X,t}, \epsilon_{Y,t}$ are iid zero mean white noise.

For a VAR model:

$$f_{jk,i}(x) = \beta_{jk,i}x, \quad \beta_{jk,i} \in \mathbb{R}$$

If we have a ground truth of $X \rightarrow Y, Y \not\rightarrow X$ with maximal lag p , then $f_{12,i}(\cdot) = 0$ for all $i = 1, \dots, q$ and $f_{21,i}(\cdot) = 0$ for all $i > p$.

These models are ideal due to their flexibility and ease of manipulation for deriving mathematical properties. A drawback is whilst the independence of white noise terms is helpful for mathematical analyses, a conditional variance structure may give us further flexibility to improve our test's relative power to the existing methods.

S1.3 GARCH

A univariate generalised autoregressive conditional heteroscedasticity (GARCH) noise model is of the form:

$$\begin{aligned} \epsilon_{i,t} &= \sqrt{h_{i,t}} \cdot z_{i,t} \\ h_{i,t} &= \alpha_0 + \alpha_1 \epsilon_{i,t-1}^2 + \beta_1 h_{i,t-1} \end{aligned}$$

where $z_{i,t}$ is some iid heavy-tailed random variable. We can then modify the VAR framework to incorporate this noise in (could equally be done with NAAR):

$$X_t = \epsilon_{x,t}, \quad Y_t = \phi Y_{t-1} + \beta X_{t-2} + \epsilon_{y,t}$$

This model has shown positive signs in preliminary testing phase against the asymptotic measures as well as the established MBB method, but will prove more difficult to derive an analytical distribution of the test statistic. Additionally, the models which have shown promise do not satisfy the strong mixing requirement that will likely be necessary.

S1.4 Preliminary Results

We will illustrate the performance of one model which has shown strength in preliminary testing. Adapted[†] from Yin et al. [11], the following is a NAAR(3) model:

$$X_t = 0.5X_{t-1} + \epsilon_t^X, \quad Y_t = 0.5Y_{t-1} + \text{sgn}(X_{t-3})|X_{t-3}|^{\frac{3}{4}} + \epsilon_t^Y, \quad \epsilon^X \stackrel{\text{iid}}{\sim} t_{10}, \quad \epsilon^Y \stackrel{\text{iid}}{\sim} t_2$$

We generate X, Y with length $n = 100$, and calculate the GC F -test coefficients and p-values for both the $X \rightarrow Y$ direction and $Y \rightarrow X$ direction. We then perform the MBB (no shift) hypothesis test with $B = 200$, and the shifted MBB test with the same B . This is replicated $N = 500$ times and we record the Type I error and the power of these tests as we know the ground truth of $X \rightarrow Y$. From Table 2 we can observe that the shifted bootstrap test dominates both the bootstrap test and F -test in reduction of Type I error, and only a marginal reduction in power compared to the F -test. Source code can be viewed [here](#).

[†]Model 7 in paper: we removed the extremal dependence term as original causal coefficient included $\mathbb{I}[X_{t-3} > \mu_X]$. Since we are now including all values we have to account for negative X_{t-3} values, which is why we have added the additional sign and absolute terms.

Table 2: Preliminary results from NAAR(3) model in GC inference

Procedure	Power	Size
<i>F</i> -test	0.891	0.047
Bootstrap (No Shift)	0.270	0.030
Bootstrap (Shift)	0.792	0.012

S1.5 Structural Causality

An alternative project goal that was considered was the shifted MBB test’s application in structural causality, drawing on methods such as the PC algorithm [14] using the structural causal model framework detailed by Pearl [15]. Though this would be a fascinating direction, the literature in optimising these algorithms is incredibly rich and it seems unfeasible that this algorithm will outperform the current work being done on optimising the PC algorithm (see Jakob Runge’s Causal Inference Lab `tigramite` Python package [16] and their method comparison [17]). The scope is somewhat challenging considering the timeframe given for this project so this direction can be considered retired.

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